# PHAM HA KHANH CHI

Computer Science · Mathematics · Artificial Intelligence
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## EDUCATION

**British Vietnamese International School Hanoi** | *Hanoi, Vietnam* | *2018–2027*
- A Levels (2025–2027): 4 A* predicted — Mathematics, Further Mathematics, Physics and English Literature.
- IGCSE (2023–2025): 9 A* and 1 A.

## QUALIFICATIONS

**ACT** | *2026*
Composite 35; English 35; Math 35; Reading 35.

**IELTS UKVI Academic** | *2026*
Overall 8.0; Listening 9.0, Reading 8.0, Speaking 7.0, Writing 7.0.

**HSK Level 4 — Chinese** | *2026*
270/300.

**Cambridge IGCSE** | *2025*
9 A* and 1 A

**Cambridge A Level** | *2026*
A* in Mathematics.

## AWARDS & COMPETITIONS

**Academic Scholarship** | *BVIS Hanoi* | *2026–2027*
Awarded in recognition of outstanding academic performance.

**Northern Vietnam Young Informatics Competition** | *Regional* | *2025*
Third Prize.

**Central Vietnam and Central Highlands Olympiad — AI Challenge** | *National* | *2025*
Consolation Prize; progressed from online preliminary rounds to the in-person final.

**UK Mathematics Trust — Senior Mathematical Challenge** | *International* | *2025*
Silver Certificate.

**UK Mathematics Trust — Intermediate Mathematical Challenge** | *International* | *2024*
Gold Certificate.

**Cambridge Upper Secondary Science Competition** | *International* | *2023–2024*
Gold Prize for a team project investigating storage temperature and microbial growth in melons.

**FOBISIA** | *International* | *2022*
Individual Silver in the 200m run and Bronze in the 300m; team Bronze in Girls' Football.

**BVIS Academic Attainment & Learning Attitude Certificates** | *School* | *2023–2026*
Repeated recognition across subjects and terms, including Further Mathematics, Physics, Mathematics, Computer Science, Business Studies, English Literature and Vietnamese.

## RESEARCH & PROFESSIONAL DEVELOPMENT

**Rethinking Attention Sinks in Qwen3-1.7B** | *Independent research* | *Feb–Sep 2026* | *4 hours/week × 36 weeks*
- Investigated why certain attention heads in a transformer language model assign unusually high attention to the initial token.
- Used grouped-query attention as a natural experiment; findings indicate that the query pathway is a causal contributor to sink behaviour.

**Echo Chambers in Recommender Systems** | *STEM Club research* | *Apr 2026–present*
- Research mentor for a simulation-based investigation of whether recommender algorithms narrow users' content exposure.
- Helped the team identify bugs behind misleading early results and redesign the simulation so it could test for a genuine effect; the team is applying for the CREST Silver Award.

**Storage Temperature and Microbial Growth in Melons** | *Team research* | *2023–2024* | *2 hours/week × 24 weeks*
- Compared microbial growth across three melon varieties stored at four temperatures.
- Contributed to the project awarded Gold in the Cambridge Upper Secondary Science Competition 2023–2024.

## PROJECTS

**Soul Garden** | *iOS app · AI-assisted wellbeing* | *Feb 2025–present* | *4 hours/week × 40 weeks*
- Developed an app combining an AI chatbot with a journal and a calming virtual garden.

**Student Mentoring Website** | *School initiative* | *In progress* | *0.5 hour/week × 8 weeks*
- Collaborating with an Academic Prefect on a platform to connect younger students with approved older mentors by subject, with shared notes, study resources and subject group chats.

**Interactive Friendship Game** | *School initiative* | *In progress* | *1 hour/week*
- Leading members of school's Tech committees to collaborate with the school's well-being counsellor to develop a storyline, interactive game for new or needed students to role play a character navigating social interactions in school setting; the goal is to teach students how to handle similar real-life scenarios.

## LEADERSHIP & SERVICE

**Co-founder, STEM Club** | *BVIS Hanoi* | *2025–present* | *1 hour/week × 20 weeks*
- Co-founded a twenty-three-member club that gives younger students hands-on experience of scientific research through student-led teams.
- Support the research process from forming hypotheses and designing experiments to evaluating limitations, interpreting results and writing reports.

**EDIB Prefect** *BVIS Hanoi* | *2025–present*
- Helped design a PSHE lesson on equality, diversity, inclusion and belonging using discussion and role-play.
- Planning an emotional-awareness and regulation ECA informed by consultation with the school counsellor; the activity will be launched in term 2a

**House Captain, Hanoi House** | *BVIS Hanoi* | *2025–2026*
- Progressed through deputy house leadership roles in Years 9 and 10 before becoming House Captain in Year 12.
- Helped coordinate a school-assembly performance, Christmas market and team-building activities; introduced weekly quizzes.
- Led Hanoi House to House Point Champion in Primary and second place in Secondary.

**NEXUS — High School Innovation Challenge** | *Organising team, Academic Department* | *2026*
- Designed a sponsorship profile, pitched to Code Dream and secured its commitment as the competition's academic sponsor.
- Grew it to 50 teams national-wide with $4K in sponsorship; broadcast on national TV news.

**Volunteer at AN Early Intervention Center** | *2026* | *4 hours/week × 10 weeks*
- Cared for children at kindergarten school for special kids with delay in development

## PROGRAMMES & ADDITIONAL LEARNING

**CS50's Introduction to Artificial Intelligence with Python** | *HarvardX* | *2026* | *International* | *3 hours/week × 20 weeks*
Completed the course, including all twelve projects.

**AI Programming for Beginners** | *ICT for People Institute · 1001 School* | *National* | *Jul 2025*
High Distinction.

**Introduction to AI** | *ICT for People Institute · 1001 School* | *National* | *Sep–Nov 2024*
High Distinction.

**Duke of Edinburgh's International Award** | *2023–2025* | *International* | *1 hour/week × 24 weeks*
Bronze and Silver Awards; Silver achieved 8 December 2025.

## ADDITIONAL ACTIVITIES & SKILLS

**Performing Arts** | *BVIS Hanoi* | *2022–2024* | *2 hours/week × 10 weeks*
- Performed in three consecutive school musicals: Storyteller in Shrek the Musical (2022), Hilda in Rock of Ages (2023), and Zeke / Storyteller in The Wizard of Oz (2024).
- Supported younger cast members in their first production during The Wizard of Oz.

**Taekwondo** | *2023–2026* | *1 hour/week × 40 weeks*
Red belt, 1st Geup

**Junior Varsity Girls' Football Team** | *2022–2023*

**Junior Varsity Girls' Volleyball Team** | *2022–2023*

**Guitar** | *Young Hit young Beat*| *2023–2025* | *2 hours/week × 20 weeks*

**Technical Skills**
Python · Java

**Languages**
Vietnamese (native) · English (IELTS 8.0) · Chinese (HSK 4, 270/300)